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Editor-in-Chief
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Dear Editor-in-Chief,

Please find enclosed the manuscript entitled “**Distilling Evolutionary Optimization into Compact Language Models for Real-Time UAV Relay Positioning in Vehicular THz Networks**” for consideration as a Regular Paper in *IEEE Transactions on Vehicular Technology*.

Scope and motivation. Sixth-generation vehicular backhaul at 300 GHz demands UAV relay positioning that adapts to evolving mission contexts—obstacle maps, weather, multi-site topologies—within sub-second latency budgets. Evolutionary solvers deliver near-optimal placements but are too slow for real-time use; fixed-feature regressors are fast but cannot absorb new input modalities without redesign. This paper bridges the gap through *optimizer distillation*: differential-evolution solutions are distilled into TinyLlama-1.1B via QLoRA. The resulting text-in/text-out interface provides *input extensibility*—new modalities are added as prompt sentences with lightweight fine-tuning, requiring no architectural changes.

Key contributions.

1. **Input extensibility:** adding building obstructions requires only 500 labels and 16 min of fine-tuning; the language model then surpasses the blind MLP (142.6 vs. 137.8 Mbps).
2. **Comprehensive evaluation:** 100 scenarios, nine baselines (MLP, DeepSets, PPO, SAC, TD3, heuristics), with Bonferroni-corrected statistics, bootstrap CIs, and effect sizes.
3. **Real-time inference:** LoRA merging cuts latency from 10.4 s to 822 ms; TensorRT targets 150–200 ms for vehicular edge deployment.
4. **Hybrid mobility:** centroid tracking with periodic VLA overrides sustains near-optimizer throughput at 0–120 km/h highway speeds.
5. **Energy efficiency:** 13.2% throughput-per-Joule gain over centroid heuristics, with log-linear data scaling confirmed by ablation.

Relevance to TVT. The manuscript addresses vehicular communication optimization, UAV-assisted relay networking, and THz backhaul—core topics of *IEEE Transactions on Vehicular Technology*. The hybrid mobility experiments at 0–120 km/h directly target highway V2X deployment scenarios.

Declarations. This manuscript has not been published elsewhere and is not under consideration by any other journal. Evaluation code and trained models will be released upon acceptance. There are no conflicts of interest to declare.

Respectfully submitted,

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